



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Example code:

```
Users > sumit > Documents > java > J FinalKeywordDemo.java > College
1  final class College
2  {
3      void displayCollege()
4      {
5          System.out.println(x: "College Name: SIT Pune");
6      }
7  }
8
9  // Parent class
10 class Animal
11 {
12     // Final method
13     final void sound()
14     {
15         System.out.println(x: "Animals make different sound");
16     }
17 }
18
19 // Child class
20 class Dog extends Animal
21 {
22     void displayDog()
23     {
24         System.out.println(x: "Dog barks");
25     }
```

```

26     // Uncommenting this will give a compilation error
27     /*
28     void sound()
29     {
30         System.out.println("Dog barks");
31     }
32     */
33 }
34 public class FinalKeywordDemo
35 {
    Run | Debug
36     public static void main(String[] args)
37     {
38         // Final variable
39         final int MAX_MARKS = 100;
40
41         System.out.println("Maximum Marks = " + MAX_MARKS);
42
43         Dog d = new Dog();
44         d.sound();           // Inherited final method
45         d.displayDog();
46
47         College c = new College();
48         c.displayCollege();
49     }
50 }

```

Output:

```

java — zsh — 161x46
Last login: Wed Jul 29 12:32:14 on ttys000
sumit@Sumit ~ % cd documents
sumit@Sumit documents % cd java
sumit@Sumit java % javac FinalKeywordDemo.java
sumit@Sumit java % ./FinalKeywordDemo
zsh: no such file or directory: ./FinalKeywordDemo
sumit@Sumit java % java FinalKeywordDemo
Maximum Marks = 100
Animals make different sound
Dog barks
College Name: SIT Pune
sumit@Sumit java % date
Thu Jul 30 11:45:33 IST 2026
sumit@Sumit java % █

```

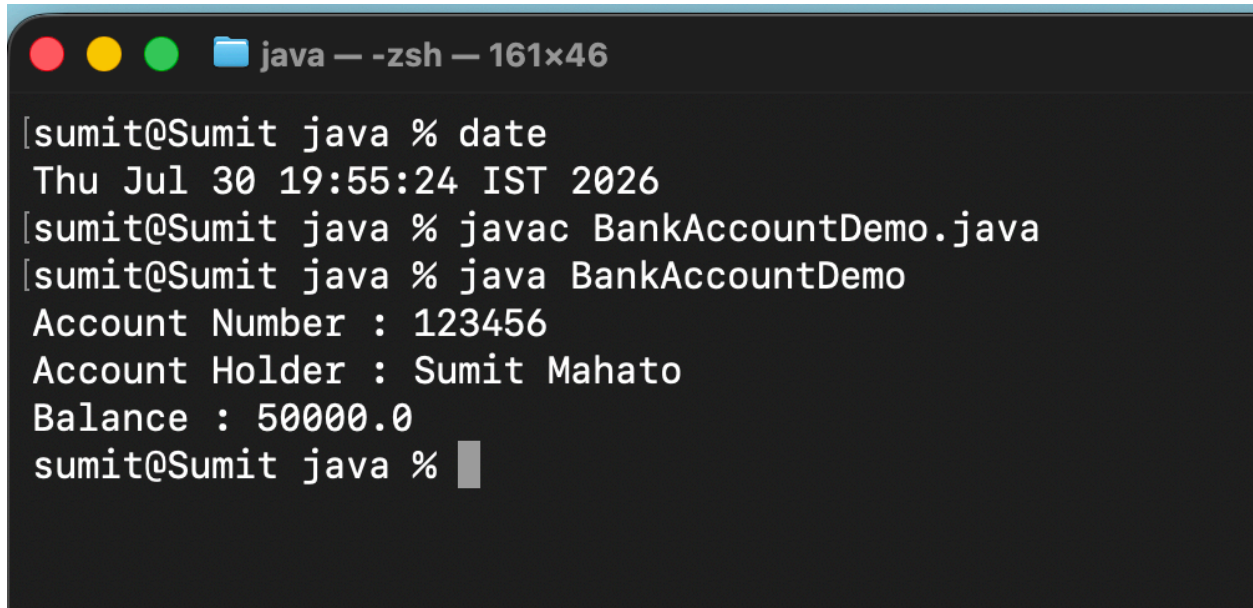
Exercise

1. Create a **Bank Account program** where account number is final and cannot be changed.

Source Code:

```
Users > sumit > Documents > java > J class BankAccount.java > BankAccountDemo
1  class BankAccount
2  {
3      final int accountNumber;
4      String accountHolder;
5      double balance;
6
7      BankAccount(int accountNumber, String accountHolder, double balance)
8      {
9          this.accountNumber = accountNumber;
10         this.accountHolder = accountHolder;
11         this.balance = balance;
12     }
13
14     void display()
15     {
16         System.out.println("Account Number : " + accountNumber);
17         System.out.println("Account Holder : " + accountHolder);
18         System.out.println("Balance : " + balance);
19     }
20 }
21
22 public class BankAccountDemo
23 {
24     public static void main(String[] args)
25     {
26         BankAccount b = new BankAccount(123456, "Sumit Mahato", 50000);
27
28         b.display();
29     }
30 }
```

Output:

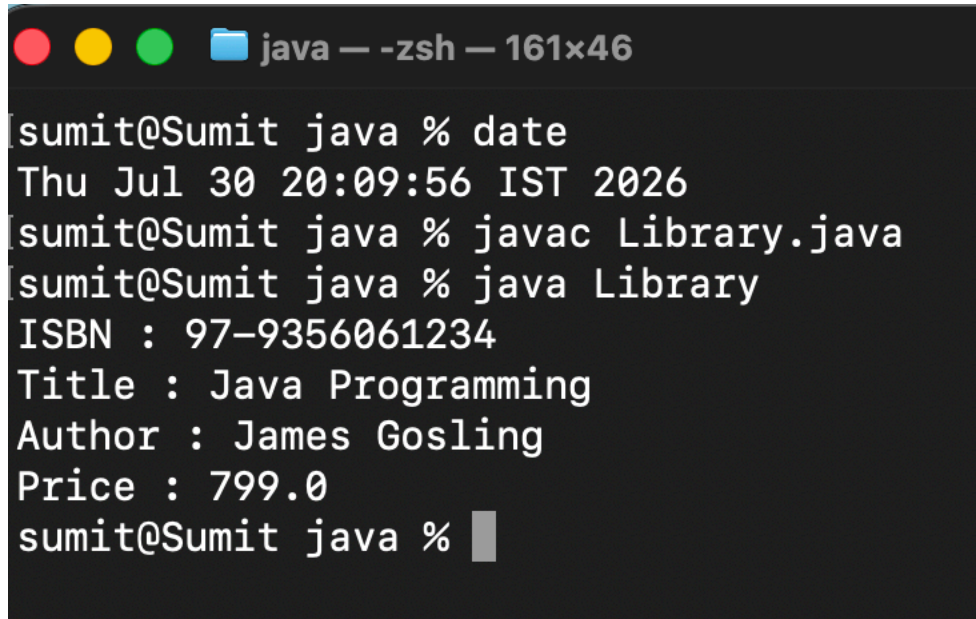
A terminal window with a dark background and a light blue title bar. The title bar contains three colored circles (red, yellow, green) and a folder icon, followed by the text "java — -zsh — 161x46". The terminal shows the following text:

```
[sumit@Sumit java % date  
Thu Jul 30 19:55:24 IST 2026  
[sumit@Sumit java % javac BankAccountDemo.java  
[sumit@Sumit java % java BankAccountDemo  
Account Number : 123456  
Account Holder : Sumit Mahato  
Balance : 50000.0  
sumit@Sumit java %
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Source Code:

```
Users > sumit > Documents > java > Library.java > Library > main(String[])
1  class Book
2  {
3      final String isbn;
4      String title;
5      String author;
6      double price;
7
8      Book(String isbn, String title, String author, double price)
9      {
10         this.isbn = isbn;
11         this.title = title;
12         this.author = author;
13         this.price = price;
14     }
15
16     void display()
17     {
18         System.out.println("ISBN : " + isbn);
19         System.out.println("Title : " + title);
20         System.out.println("Author : " + author);
21         System.out.println("Price : " + price);
22     }
23 }
24
25 public class Library
26 {
27     Run | Debug
28     public static void main(String[] args)
29     {
30         Book b = new Book("97-9356061234", "Java Programming", "James Gosling", 799);
31
32         b.display();
33     }
34 }
35 }
```

Output:A terminal window with a dark background and light-colored text. The window title bar shows three colored circles (red, yellow, green) and a folder icon, followed by the text 'java — -zsh — 161x46'. The terminal content shows a series of commands and their outputs. The first command is 'date', which outputs 'Thu Jul 30 20:09:56 IST 2026'. The second command is 'javac Library.java'. The third command is 'java Library', which outputs 'ISBN : 97-9356061234', 'Title : Java Programming', 'Author : James Gosling', and 'Price : 799.0'. The prompt 'sumit@Sumit java %' is shown at the end of the last line, followed by a cursor.

```
sumit@Sumit java % date
Thu Jul 30 20:09:56 IST 2026
sumit@Sumit java % javac Library.java
sumit@Sumit java % java Library
ISBN : 97-9356061234
Title : Java Programming
Author : James Gosling
Price : 799.0
sumit@Sumit java %
```

Conclusion:

The program successfully demonstrated the use of the final keyword in Java. A final variable cannot be reassigned after initialization, a final method cannot be overridden by a subclass, and a final class cannot be inherited. Thus, the final keyword helps protect data, prevents unwanted modifications, and improves the security and reliability of Java programs.